

Program	Diploma Engineering (DE)	Semester - 3
Type of Course	Engineering Science	
Prerequisite	Basic knowledge of the science and mathematics	
Course Objective	To acquire conceptual knowledge of digital electronics & simple electronics components and circuits.	

Teaching Scheme (Contact Hours)				Examination Scheme				
Lecture	Tutorial	Lab	Credit	Theory Marks		Practical Marks		Total Marks
				SEE	CIA	SEE	CIA	
2	2	0	4	40	30	20	10	100

SEE - Semester End Examination, CIA - Continuous Internal Assessment (It consists of Assignments/Seminars/Presentations/MCQ Tests, etc.)

Course Content		T - Teaching Hours W - Weightage	
Sr.	Topics	T	W
1	Binary Systems Introduction of Digital Computers and Digital Systems, Base Conversion - Binary, Decimal, Octal, Hexadecimal, Types of Codes, Binary Addition, Binary Subtraction	5	20
2	Binary Logic And Boolean Algebra Logic Gates, Implementation of Logic Gates using Universal Gates, Postulates, Laws of Boolean Algebra, Boolean Algebra Theorem - De-Morgan's Theorems, Simplification of Boolean Expression, Draw Logic Circuit for given Boolean Expression, Write Boolean Expression for given Logic Circuit	5	20
3	Boolean Function Implementation Boolean Function and Boolean Forms, Methods of Constructing Karnaugh Map (K-Map)- 2-Variable K-map, 3-Variable K-map, 4-Variable K-map, K-Map using Don't Care Condition	7	20
4	Basic Combinational Logic Adder - Half Adder, Full Adder, Subtractor - Half Subtractor, Full Subtractor, Magnitude Comparator, Decoder - 2 to 4 Decoder, 3 to 8 Decoder, Multiplexer - 2 to 1 multiplexer, 4 to 1 multiplexer	7	20
5	Memories and Programmable Logic Devices Memory Organization and Operation, Classification of Memory-ROM, RAM, ROM as PLD, Programmable Array Logic (PAL)	6	20
Total		30	100

Suggested Distribution Of Theory Marks Using Bloom's Taxonomy						
Level	Remembrance	Understanding	Application	Analyze	Evaluate	Create
Weightage	20	40	40	0	0	0

NOTE : This specification table shall be treated as a general guideline for the students and the teachers. The actual distribution of marks in the question paper may vary slightly from above table.

Course Outcomes

At the end of this course, students will be able to:

C01	practice number system conversions, error detection & reflected code.
C02	implement working of logic gates, boolean algebra laws and De-Morgan's theorems.
C03	perform boolean function simplification.
C04	design combinational logic circuits using adder, subtractor, decoder, multiplexer.
C05	articulate types of memory, programmable logic array and programmable array logic.

Reference Books

1.	Digital Logic and Computer Design By Mano M. Morris Pearson publication
2.	Digital Electronics Principles By Malvino & Leech Tata McGraw-Hills publication
3.	Electronics Principles By Albert Paul Malvino Tata McGraw-Hills
4.	Fundamentals of Digital circuits By A. Anand Kumar PHI

List of Tutorial

1.	Number System Conversion - Decimal to Binary (A) $(64)_{10} = (\text{_____})_2$ $(92.29)_{10} = (\text{_____})_2$ $(26.62)_{10} = (\text{_____})_2$ - Decimal to Octal (A) $(64)_{10} = (\text{_____})_8$ $(92.29)_{10} = (\text{_____})_8$ $(26.62)_{10} = (\text{_____})_8$ - Decimal to Hexadecimal (A) $(64)_{10} = (\text{_____})_{16}$ $(92.29)_{10} = (\text{_____})_{16}$ $(26.62)_{10} = (\text{_____})_{16}$ - Decimal to Binary (B) $(553.652)_{10} = (\text{_____})_2$ $(368.772)_{10} = (\text{_____})_2$ $(543.001)_{10} = (\text{_____})_2$ - Decimal to Octal (B) $(553.652)_{10} = (\text{_____})_8$ $(368.772)_{10} = (\text{_____})_8$ $(543.001)_{10} = (\text{_____})_8$ - Decimal to Hexadecimal (B) $(553.652)_{10} = (\text{_____})_{16}$ $(368.772)_{10} = (\text{_____})_{16}$ $(543.001)_{10} = (\text{_____})_{16}$
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7. Given that $(4096)_{10} = (1000)_x$, Find the value of x . (C)

2. Number System Conversion-Binary to Decimal, Octal, Hexadecimal

1. Binary to Decimal (A)
 - $(111011)_2 = (\rule{1cm}{0.4pt})_{10}$
 - $(1110111.010)_2 = (\rule{1cm}{0.4pt})_{10}$
 - $(1101011.111)_2 = (\rule{1cm}{0.4pt})_{10}$
2. Binary to Octal (A)
 - $(111011)_2 = (\rule{1cm}{0.4pt})_8$
 - $(1110111.010)_2 = (\rule{1cm}{0.4pt})_8$
 - $(1101011.111)_2 = (\rule{1cm}{0.4pt})_8$
3. Binary to Hexadecimal (A)
 - $(111011)_2 = (\rule{1cm}{0.4pt})_{16}$
 - $(1110111.010)_2 = (\rule{1cm}{0.4pt})_{16}$
 - $(1101011.111)_2 = (\rule{1cm}{0.4pt})_{16}$
4. Binary to Decimal (B)
 - $(111011.11011)_2 = (\rule{1cm}{0.4pt})_{10}$
 - $(1110111.0011)_2 = (\rule{1cm}{0.4pt})_{10}$
 - $(11101100.000110)_2 = (\rule{1cm}{0.4pt})_{10}$
5. Binary to Octal (B)
 - $(111011.11011)_2 = (\rule{1cm}{0.4pt})_8$
 - $(1110111.0011)_2 = (\rule{1cm}{0.4pt})_8$
 - $(11101100.000110)_2 = (\rule{1cm}{0.4pt})_8$
6. Binary to Hexadecimal (B)
 - $(111011.11011)_2 = (\rule{1cm}{0.4pt})_{16}$
 - $(1110111.0011)_2 = (\rule{1cm}{0.4pt})_{16}$
 - $(11101100.000110)_2 = (\rule{1cm}{0.4pt})_{16}$
7. Convert the decimal number 250.5 to base 3, base 4, base 7 & base 8 (C)

3. Number System Conversion-Octal to Binary, Decimal, Hexadecimal

1. Octal to Decimal (A)
 - $(45)_8 = (\rule{1cm}{0.4pt})_{10}$
 - $(66.34)_8 = (\rule{1cm}{0.4pt})_{10}$
 - $(51.29)_8 = (\rule{1cm}{0.4pt})_{10}$
2. Octal to Binary (A)
 - $(45)_8 = (\rule{1cm}{0.4pt})_2$
 - $(66.34)_8 = (\rule{1cm}{0.4pt})_2$
 - $(51.29)_8 = (\rule{1cm}{0.4pt})_2$
3. Octal to Hexadecimal (A)
 - $(45)_8 = (\rule{1cm}{0.4pt})_{16}$
 - $(66.34)_8 = (\rule{1cm}{0.4pt})_{16}$
 - $(51.29)_8 = (\rule{1cm}{0.4pt})_{16}$
4. Octal to Decimal (B)
 - $(553.652)_8 = (\rule{1cm}{0.4pt})_{10}$
 - $(368.772)_8 = (\rule{1cm}{0.4pt})_{10}$
 - $(543.001)_8 = (\rule{1cm}{0.4pt})_{10}$
5. Octal to Binary (B)
 - $(553.652)_8 = (\rule{1cm}{0.4pt})_2$
 - $(368.772)_8 = (\rule{1cm}{0.4pt})_2$
 - $(543.001)_8 = (\rule{1cm}{0.4pt})_2$

6. Octal to Hexadecimal (B)
 $(553.652)_8 = (\rule{1cm}{0.4pt})_{16}$
 $(368.772)_8 = (\rule{1cm}{0.4pt})_{16}$
 $(543.001)_8 = (\rule{1cm}{0.4pt})_{16}$
7. Covert Number $(4433)_5$ to Base 10 and Base 2 (C)

4. Number System Conversion-Hexadeciam to Binary, Octal, Decimal

1. Hexadecimal to Decimal (A)
 $(A4)_{16} = (\rule{1cm}{0.4pt})_{10}$
 $(B6D.34)_{16} = (\rule{1cm}{0.4pt})_{10}$
 $(AE4.40)_{16} = (\rule{1cm}{0.4pt})_{10}$
2. Hexadecimal to Binary (A)
 $(A4)_{16} = (\rule{1cm}{0.4pt})_2$
 $(B6D.34)_{16} = (\rule{1cm}{0.4pt})_2$
 $(AE4.40)_{16} = (\rule{1cm}{0.4pt})_2$
3. Hexadecimal to Octal (A)
 $(A4)_{16} = (\rule{1cm}{0.4pt})_8$
 $(B6D.34)_{16} = (\rule{1cm}{0.4pt})_8$
 $(AE4.40)_{16} = (\rule{1cm}{0.4pt})_8$
4. Hexadecimal to Decimal (B)
 $(553.652)_{16} = (\rule{1cm}{0.4pt})_{10}$
 $(368.772)_{16} = (\rule{1cm}{0.4pt})_{10}$
 $(543.001)_{16} = (\rule{1cm}{0.4pt})_{10}$
5. Hexadecimal to Binary (B)
 $(553.652)_{16} = (\rule{1cm}{0.4pt})_2$
 $(368.772)_{16} = (\rule{1cm}{0.4pt})_2$
 $(543.001)_{16} = (\rule{1cm}{0.4pt})_2$
6. Hexadecimal to Octal (B)
 $(553.652)_{16} = (\rule{1cm}{0.4pt})_8$
 $(368.772)_{16} = (\rule{1cm}{0.4pt})_8$
 $(543.001)_{16} = (\rule{1cm}{0.4pt})_8$
7. Convert the decimal number $(2598.675)_{10}$ to Base 2, Base 4, Base 8, Base 16. (C)

5. Error Detection Techniques, Decimal to BCD, Binary Addition, Binary Substraction

1. Write down a table of 0 to 7 decimal numbers into binary form and mention odd parity for all numbers. (A)
2. Write down a table of 50 to 55 decimal numbers into binary form and mention even parity for all numbers. (A)
3. Decimal to BCD (A)
 $(3441)_{10} = (\rule{1cm}{0.4pt})_{BCD}$
 $(2469)_{10} = (\rule{1cm}{0.4pt})_{BCD}$
 $(1895)_{10} = (\rule{1cm}{0.4pt})_{BCD}$
4. Find the XS-3 code of 37 (A)
5. Perform the following (A)
 $1101111 + 110111$
 $11101111 + 110111$
 $11011 - 1101$
 $1110111 - 11011$

	6. Perform the following (C) $111011.101 + 1111.11$ $110111.110 + 111.101$ $1010.010 - 111.111$ $111011.11 - 11110.01$
6.	Logic Gate Trainer Kit Perform hands on practical of Logic gates in Digital IC Logic Trainer Kit.
7.	Boolean Expression Simplification-Set-1 - Simplify following expression $B'C + AB + BC$ (A) (A) $A(A' + C) (A'B + C')$ (B) - Prove that following expression $A + A'B + AB' = A + B$ (A) $(AB + BC + CA)' = A'B' + B'C' + C'A'$ (B) $((A + B)' + A'B' + (A' + B))'$ (C)
8.	Boolean Expression Simplification-Set-2 - Prove that following expression: $AB + BC + A'C = AB + A'C$ (A) - Show that following expression: $AB'C + B + BD' + ABD' + A'C = B + C$ (A) $(A + C) (A + D) (B + C) (B + D) = AB + CD$ (B) $AB + AB'C + BC' = AC + BC'$ (C) - Simplify following expression: $F = (ABC)' + (AB)'C + A'BC' + A(BC)' + AB'C$ (A) $F(A, B, C) = (A + B) (A + C') + A'B' + A'C'$ (C)
9.	Boolean Expression Simplification using K-Map-Set-1 Simplify following expression using K-Map: $F = \sum m(0, 2, 3, 4, 5, 6)$ (A) $F(A, B, C, D) = \sum m(0, 1, 2, 5, 8, 9, 10)$ (A) $F(A, B, C, D) = \prod M(2, 8, 9, 10, 11, 12, 14)$ (A) $F(P, Q, R, S) = \prod M(0, 1, 3, 4, 5, 6, 7, 13, 15)$ (B) $F(A, B, C, D) = \sum m(0, 2, 4, 6, 7, 8, 10, 12, 13, 15)$ (B)
10.	Boolean Expression Simplification using K-Map-Set-2 $F(A, B, C, D) = \sum m(1, 2, 4, 6, 7, 11, 15) + d(0, 3)$ (A) $F(W, X, Y, Z) = \sum m(1, 3, 5, 8, 9, 11, 15) + d(2, 13)$ (A) - Reduce the expression in SOP and POS form using K-Map: $F(A, B, C, D) = \sum m(1, 5, 6, 12, 13, 14) + d(2, 4)$ (B) - Reduce using K-Map: $F(A, B, C, D) = \sum m(6, 7, 8, 9) + d(10, 11, 12, 13, 14, 15)$ (C)
11.	Boolean expression simplification using K-map and Draw circuit using A-O-I gate $Y = F(W, X, Y, Z) = \sum m(0, 2, 5, 7, 8, 10, 12, 15)$ (A) $Y = F(A, B, C, D) = \sum m(0, 1, 2, 4, 6, 7, 8, 9, 11, 12, 15) + d(3, 5, 10, 14)$ (B) $Y = F(W, X, Y, Z) = \prod M(0, 1, 2, 6, 7, 8, 11, 12, 15)$ (A)
12.	Boolean expression simplification using K-map and Draw circuit using A-O-I, NAND and NOR gate - Using A-O-I and NAND gate: $F = ABD + A'C'D' + A'B + A'CD' + AB'D$ (A) $F(A, B, C, D) = ABC + A'BC' + CD + AB'$ (B) - Using A-O-I and NOR gate:

	$F = (A + B' + D) (A' + C' + D') (A' + B + C) (A + C + D') (A' + C' + D) \text{ (A)}$ $F(A, B, C, D) = (A + B + C) (A' + B + D') (A + C + D) (A' + B') \text{ (B)}$
13.	Combinational Circuits - Implement 3x8 decoder using 2x4 decoder. (A) - Implement 4x16 decoder using 3x8 decoder. (C) - Implement following Boolean function using 8 : 1 multiplexer: $Y(A, B, C, D) = \sum m(15, 7, 4, 6, 8, 9, 12, 14) \text{ (A)}$ $F(A, B, C, D) = \sum m(2, 3, 5, 7, 8, 9, 12, 13, 14, 15) \text{ (B)}$ - Implement following Boolean function using 4 : 1 multiplexer: $F(A, B, C) = \sum m(1, 3, 5, 6) \text{ (A)}$ $F(A, B, C) = \sum m(1, 2, 4, 7) \text{ (B)}$